

On Building the Largest and Cross-Linguistic Turkish Dependency Corpus

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Abstract—In this paper, we aim to introduce the dependency annotation process of the largest and the only cross-linguistic Turkish dependency treebank which was translated from the original Penn Treebank corpus. Within the scope of this project, 16,400 sentences have been morphologically and semantically annotated, and the dependency relations were manually carried out by a team of linguists. It is hoped that this project will serve as a base for a successful dependency parser and a system which can automatically perform the bi-directional conversion between constituency and dependency trees.

Index Terms—Dependency, TreeBank, Dependency Parsing

I. INTRODUCTION

Natural Language Processing (NLP) has been a popular field developing with the contributions of computer science, computational linguistics, and machine learning studies. Dependency parsing is one of the main components of NLP. It is used in all kinds of speech applications, dictionaries, and so on. Previous studies showed that dependency parsers didn't work well when they were applied to languages that they are not compatible with. Therefore, there is an increasing need of cross-linguistic treebanks to enhance the performance of dependency parsers. There are several other Turkish treebanks created as an attempt to present the Turkish language in the Universal Dependency (UD) framework [1]–[4]. These treebanks served well to present the Turkish language and create a base-line to designate a standard annotation guideline for the following works. However, some of them either lacked adequate data, linguistic resources or a standard annotation scheme.

In this paper, we introduce the Turkish Dependency Treebank (TDT) created by a team of linguists, following the guidelines of the UD and Stanford Universals. Being the largest dependency treebank in Turkish, TDT includes a total of 16,400 syntactically, morphologically and semantically annotated sentences. Another unique facility of TDT is that the corpus is taken from one of the most standard treebanks of English, the Penn Treebank, and then translated into Turkish,

the sentences are tagged both in English and in Turkish. A manual transition was needed because of the different branching typology of the two languages. This type of cross-linguistic treebanks exist only with a limited corpora. Such a large corpus manually annotated in two languages gives us an impeccable overview of what the dependency trees look like in two different languages with efficient data while being the first example of a Turkish-English cross-linguistic treebank. This type of treebanks also help improve the developing technologies made possible with NLP studies, such as machine translation.

The ultimate goal of this study is to obtain a parser in the future that can automatically parse the sentences and converse dependency relations into constituency trees and another system which can turn constituency trees into dependency.

This paper consists of a literature review section where we outline the dependency approaches and the previous treebank studies followed by a data and methodology section that includes statistical information of our data and the linguistic approach we adopted during the annotation process. Then, there is a section where we explain the inconsistencies in our annotations. Lastly, there is a conclusion section where we shortly wrap up our contribution.

II. RELATED WORK

The use of dependency grammar and dependency syntax significantly increased in the last two decades by the virtue of the development of natural language processing (NLP) studies. The foundation of the dependency grammar, however, starts as early as the 1950's with [5]. The dependency structures were studied even before Tesnière by linguists such as [6] and [7]. Yet, these studies did not provide a linguistics theory for the use of modern needs of today's dependency concept.

Tesnière introduced the dependency trees, structural order, the concept of dependency and applied his representation concepts in a variety of languages such as French, Greek, Russian, Italian, and so on. In structural order, syntactic relations are presented in a hierarchical manner as opposed

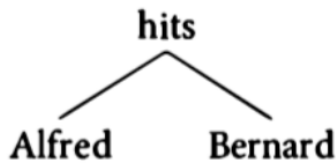


Fig. 1. The stemma for the sentence “Alfred hits Bernard”

to the linear order. He uses “stemmas” as in Figure 1 of [5] to reflect hierarchy. In Figure 1, the lines together build up the stemma for the sentence “Alfred hits Bernard”.

The predicate always comes at the top of the hierarchy and its arguments come below it. This order does not always match with the actual order of the sentences. Tesnière suggested that we have this hierarchical order in our minds and we alternate this order with the linear order during speech production.

Today, dependency grammars are divided into two in terms of the acknowledgement of this alternation. One study from the tradition that applies this distinction to the dependency grammar is the Prague Dependency Treebank (PDT) [8] developed by the Prague School of Functional and Structural Linguistics.

A famous example for the other school of thought that displays linear order is the Penn Treebank [9] that functioned between the years of 1989-1996, containing seven million annotated texts from American English. Penn Treebank (PT) annotations were made in two steps. They first did an automatic annotation and then manually fine tuned the outcome.

[10] stated that the tree models of the Penn treebank and also the other treebanks in the literature [11], [12] were sensitive to different annotation styles. [10] highlighted the need for unique treebanks for languages other than English with this study.

In 2005, The Stanford Dependencies developed for the parsing of the English language and to be used in NLP studies and in Stanford Dependency Parser. Stanford Dependencies were acknowledged as the standard for the dependency analyses of English. However, the Stanford Dependency parser could not reach an adequate accuracy when it was used with other dependency schemes. In the following years, the Universal Dependency Treebank (UDT) project [13] pioneered to develop treebanks for languages other than English by transforming the Stanford dependencies into a more inclusive annotation scheme for a diverse set of languages.

The developments in the dependency treebanking made it clear that Turkish language needed a Treebank of its own. The first Turkish language dependency treebank is METU-Sabancı Turkish Treebank developed by [1] and [2]. This treebank used a corpus that consisted of 7.262 sentences and included morphological and syntactic annotations. In 2016, this treebank was revisited by [3] under the name of ITU-METU-Sabancı Treebank (IMST) to reduce the inconsistencies of its earlier version. They succeeded to reduce inconsistencies by applying a new annotation scheme. However, lack of

an adequate linguistic team and automatic transition to the UD framework resulted in incoherent annotations. Therefore, [4] introduced The Boğaziçi-ITU-METU-Sabancı Treebank (BIMST) using the same corpus. Having a linguistic team of three people, they created a new annotation scheme for IMST and manually re-annotated the data of 5.635 sentences while introducing new dependency relations that were not present in IMST. [4] also re-annotated the Turkish PUD Treebank. Turkish-PUD treebank consisted of 1000 words. Turkish-PUD is important for cross-linguistic dependency studies because it was parallel annotated with 18 other languages. However, it had linguistic problems such as distinguishing between the compound and Noun-Noun constructions [4]. Most of these problems occurred because it was annotated in the absence of a human supervisor.

III. DATA AND METHODOLOGY

The Turkish Penn Treebank corpus contains a total of 16.400 sentences, all of which were translated into Turkish from the original English Penn Treebank corpus [9]. After the translation process, the words of each sentence went under a morphological analysis which was realized with our own semi-automatic morphological analyzer [14].

The semantic annotation of the words was also done manually. Finally, 9.600 sentences¹ went under a shallow parsing phase, in which the verb, the subject and some other basic tags were attached automatically. The other 6.800 sentences² were all annotated manually. After all the preliminary work, the dependency annotation process was completed.

The dependency annotations of Turkish Penn Treebank were carried out by 7 linguists with the help of an original interface developed by our team. The interface displays the sentences in a linear way (See Figure 2). The annotator marks the dependency relations by drawing an arrow from the dependent towards the head, then selecting the corresponding tag from a window that appears. Upon this selection, the point of the arrow shows the dependent and the root of the arrow shows the head. The tag is shown below the arc of the arrow. In case of a mistake, the links can be altered by simply reestablishing the arrows over the words.

The control, correction and evaluation processes of the data take place in another window which enlists every tagged item with their corresponding tags. This enables the annotators to arrange the data in accordance with what they are searching.

For this corpus, the annotations were checked by one of the linguists of the team and certain inconsistencies were listed using the window shown in Figure 3. The ones that pose a faulty tagging were corrected and others were debated amongst the team to either lead to the change of the tag or remained as an issue of discussion. Some of these discussions are mentioned in Section V. However, the content of these debates exceed the purpose of this paper.

¹<https://github.com/olcaytaner/TurkishAnnotatedCorpus-15>

²<https://github.com/olcaytaner/TurkishAnnotatedCorpus-20>

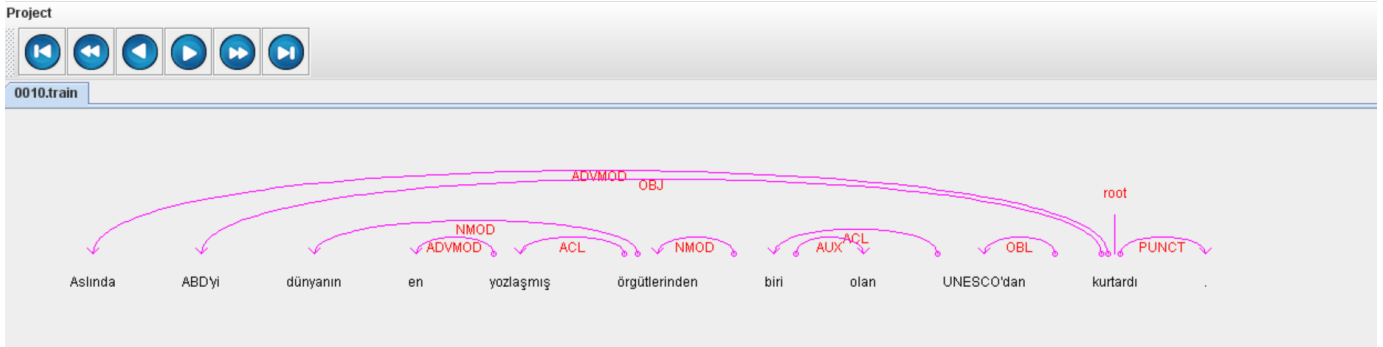


Fig. 2. The interface used to mark the dependency relations

FileName	Index	Word	Depend.	Dependency Type	Sentence
3302.train	4	mahkemeler	11	NSUBJ	Ve çalışma federal mahkemeler erken süreçte firma deneme tarihleri ayarlamak gerektiğini söyledi .
3878.train	7	mahkemeler	15	NSUBJ	Ancak, kademeli olarak, alt mahkemeler ve savcılar bu tanımlı kopma noktasına kadar zorladı .
3880.train	1	mahkemeler	9	NSUBJ	Mahkemeler ve yasama meclisleri çok farklı şekillerde karar alıyor ve her biri çok farklı hatalara
2399.train	2	MAHKEMELELER	11	NSUBJ	FEDERAL MAHKEMELELER maliyetlerini düşürmek ve sivil tetik geçikmelerini azaltmak için ilerledi .
2345.train	3	mahkemenin	11	NSUBJ	Şirket mahkemenin 27 Kasım civarında bu konuda karar vereceğini belirttiğini söyledi .
1705.train	6	mahkemenin	10	NSUBJ	Bir ABD Sprint sözcüsü " mahkemenin bunu dışı bulacağına inanıyoruz, " dedi .
1263.train	2	Mahkemesi	13	NSUBJ	Vergi Mahkemesi , " Hobisi ile ilgileniyor ancak bunu sönümlüyor, " dedi .
1244.train	3	Mahkemesi	5	NSUBJ	Ülkenin Yüksek Mahkemesi davayı reddetti , ancak Bay Garcia geçen hafta tekrar incelenmesi için
3326.train	3	Mahkemesi	13	NSUBJ	İtalyan Yüksek Mahkemesi bu yıl Parlamento'dan medya sahipliğini düzenleyen bir yasa yazmasını
3748.train	3	Mahkemesi	15	NSUBJ	Ancak Vergi Mahkemesi sınırlandırmaların makul, mantıklı ve gerçekçi olduğunu ve kesintilere izin
1026.train	4	mahkemesinin	13	NSUBJ	SciMed, temyiz mahkemesinin daha sonraki işlemler için davayı bölge mahkemesine geri
3593.train	6	mahkumiyet	12	NSUBJ	Herhangi bir tek suçlama makalesine mahkumiyet hakim Hastings görevden almak için yeterliydi .
2665.train	9	mahsulü	12	NSUBJ	Yani bir zamanlar 47 milyon ton olan patates mahsulü 35 milyona düştü .
3815.train	5	Maiorana	12	NSUBJ	Giuliani kampanyasının danışmanı Ron Maiorana , " Suç karakterde gidiyor " diyor .
3597.train	3	makalelerin	13	NSUBJ	Valhi, makalelerin Valhi ve bağı şirketlerinin Lockheed hakkındaki niyetlerini tam olarak
1551.train	14	makalesi	16	NSUBJ	Scott E. Hein, Teksas Tech Üniversitesi Lubbock, Teksas Yatırım Nispan Tahtıncı makalesi hedefi
1481.train	7	makalesi	18	NSUBJ	Christopher Cox'un 13 Ekim editör sayfa makalesi olan " " Daha Sakat Davalara Doğru " " asıl konuyu
1010.train	4	makalesi	15	NSUBJ	Bir dizi gazete makalesi , güvenilir kişiyi, Picop satışında hükümete eklik ödeme yapmakla suçladı .
1011.train	2	makam	12	NSUBJ	Yetkili makam toplam 446.5 milyon dolar tutarında kirlilik-kontrol gelir tehlihi ihraç edecek .
1610.train	10	makam	15	NSUBJ	Yollannı gönüllü olarak değiştirmeyeceklerse, belki daha büyük bir makam , kongre, bunu
2391.train	4	makbuzları	13	NSUBJ	Amerikan Jaguar depozito makbuzları 1.9 milyon ciro ile 3/8 ila 11 5/8 arttı .
2626.train	8	makbuzlarının	13	NSUBJ	Diğer aktif hisse senetleri, Amerikan mevduat makbuzlarının 1/8 ila 11 1/4 eklediği Jaguar'ı
2550.train	10	makinelere	15	NSUBJ	Star Wars çöl savagı sahnesinden gelmiş gibi görünen devasa makinelere kum tepeleri arasında
1090.train	6	makinelere	10	NSUBJ	Günümüzün bilgisayar ağlarında, bazı makinelere sık sık boğa dururken , diğerleri fazla vergi
2054.test	3	makinelere	8	NSUBJ	Elcotel, makinelere fatura verilerini toplaması, depolaması ve ilemesi için bir kredi kartı okuyucu
3301.train	5	makinelere	10	NSUBJ	O ve diğerleri, makinelere hafızayı tehlikeye atmayacak kadar zayıf olduğunu söylüyor .

Fig. 3. The window for overview of annotations

IV. MOST FREQUENT TAGS

In total, 40 different tags have been used for the annotation of the corpus. The sub-types of the tags were not included; sub-types like NMOD:poss, NMOD:tmod were grouped under the broader tag NMOD. Table I contains a list of all the tags that have been used and their frequency amongst a total of 178.121 tagged items.

Below are some of the most frequent or crucial tags which have been used for the annotation process. All of the examples have been extracted from our Turkish Penn Treebank corpus.

A. Differentiating Subjects

4 different types of subject tags have been used for the annotation: NSUBJ, NSUBJPASS, CSUBJ and CSUBJPASS.

1) **NSUBJ, NSUBJPASS**: NSUBJ marks the nominal subjects of matrix or embedded clauses which are in active voice.

(1) Kovboyılar oyunu kaybediyor .

(2) Kurulun yönlendireceği şekilde tamamladı .

In example (1) the nominal subject is a dependent of the verb of the matrix clause whereas in example (2) the nominal subject depends on the verb of the embedded clause which is itself an adjectival clause.

In clauses with passive verbs, the nominal subject is marked with NSUBJPASS both in the matrix clause (3) and in embedded clauses (4).

(3) Frankfurt'un açılışı yarım saat ertelendi .

(4) açıkların azaltılması hedeflerine ulaşılmasından

2) **CSUBJ, CSUBJPASS**: CSUBJ tag is used for the clausal subjects of active sentences or clauses. In example (5) the clausal subject "rekabetçi bir çağa girdiğimiz" is the subject of the matrix clause while in example (6) the clause "kavramsal tabanın değiştirilmesi" is the subject of an embedded clause.

(5) Rekabetçi bir çağa girdiğimiz çok açık .

(6) Kavramsal tabanın değiştirilmesi gerektiğini düşünüyorum .

In cases where the verb is passive voice, the clausal subjects are marked with CSUBJPASS. The examples (7) and (8) below show clausal subjects in matrix clauses and embedded clauses, respectively.

(7) Tesisin hangi ölçüde etkilendiği bilinmiyordu .

TABLE I
DEPENDENCY RELATIONS AND THEIR FREQUENCIES

Dependency Relation	Frequency
ACL	2.990
ADVCL	2.743
ADVMOD	12.389
AMOD	13.702
APPOS	593
AUX	1.488
CASE	5.273
CC	4.139
CCOMP	2.755
CLF	79
COMPOUND	7.859
CONJ	3.546
COP	113
CSUBJ	328
CSUBJPASS	89
DEP	69
DET	6.610
DISCOURSE	1.325
DISLOCATED	27
EXPL	1
FIXED	450
FLAT	4.103
GOESWITH	615
IOBJ	1.146
LIST	368
MARK	1.020
NEG	460
NMOD	24.856
NSUBJ	14.381
NSUBJPASS	1.356
NUMMOD	6.860
OBJ	8.717
OBL	8.175
ORPHAN	82
PARATAXIS	866
PUNCT	27.617
REPERANDUM	11
ROOT	16.318
VOCATIVE	8
XCOMP	351
Total	183.878

(8) Olumsuz etkenlerin sürmesinin beklendiğini söyledi .

B. Differentiating Arguments and Adjuncts

For arguments and adjuncts, which are dependents of the verb, 5 tags have been used: OBJ, IOBJ, CCOMP, XCOMP and OBL.

1) *OBJ*: In our analysis, the core arguments of the verb, no matter what type of case they have been assigned by the verb, fall under the category of OBJ. There are some other propositions which suggest a further division between objects with canonical (OBJ) vs. non-canonical (OBL:arg) case marking. This distinction helps separate prepositional objects from adjuncts. [15] This distinction was applied to Turkish during the re-annotation process of the Turkish IMST-UD Treebank [4]. However, considering that the syntactic function of these arguments is the same as core arguments, in our analysis they are unified under the OBJ tag.

(9) Herkes için barış zamanından bahsediyor , sonra ...

(10) konuyu haftasonu düşünmek istediğini söyledi .

As it can be seen in example (9) and (10), the accusative or ablative case marking on the verb doesn't affect the tag as long as the case is assigned by the verb and not due to semantic necessities.

2) *IOBJ*: This tag is used to mark the core objects other than the first object. Although these are usually marked with the dative case, not every dative case falls under the IOBJ category.

(11) Size 1987'den farklı olduğunu söyleyecek .

(12) Anlaşma pay sahiplerine verilmeyecektir .

3) *OBL*: This tag marks the dependents of the verbs which are not required by its argument structure, in other words the adjuncts. As can be seen in examples (13) and (14), the nouns might carry the same cases (in this case, locative and dative) as the nouns marked as OBJ or OBL. It is not the case itself that determines the tag but whether it was assigned by the verb or the semantic requirements of the sentence.

(13) Bir adam bir ofiste oturuyor .

(14) İşletme aşırı doymuş bir pazara atılmaz .

4) *CCOMP*: The CCOMP tag marks finite and non-finite clausal core arguments, which have subjects. Example (15) illustrates a finite clause marked with CCOMP, while example (16) illustrates a non-finite clause.

(15) "olumlu kanaatinden çok mutluyum" dedi .

(16) bir Japon şirketine ait olacağını öngörüyor .

5) *XCOMP*: Clausal arguments which don't have their own subjects are marked with XCOMP. In our corpus, this tag mostly corresponds to infinitival verbs marked with the suffix -mAk.

(17) Yapmak istediğimiz şey daha agresif ...

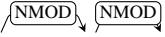
(18) yalnız 15 saniye sürdüğünü kabullenmek zor .

C. Types Of Modifiers

Another set of tags we have used in this Turkish Penn TreeBank are modifiers. There are seven distinct modifier tags used to annotate the corpus. These tags are selected by following the guidelines of Universal Stanford Dependencies [16] and considering the Turkish modifier inventory. Their functions are illustrated in the following examples from our data.

1) *NMOD*: This tag is used to designate the nominal modifiers. Nominal modifiers indicate possessive relations, part-whole relations, comparative relations, and causative relations

constructed by nominals. COMPOUND and FLAT are two other tags that are used to differentiate the compounds and proper nouns from nominal modifiers respectively.

(19)  bir sendika liderliğı türü değil .

2) *Adjectival Modifiers*: AMOD, ACL: AMOD tag marks adjectival modifiers while ACL tag marks the clausal modifiers that come before nouns. The modifying clause can be finite or non-finite. The ACL example below illustrates the participle “geçirecek” form modifying a noun.

(20)  Onu geçirecek oya sahip değil .

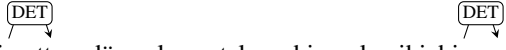
(21) “ Bu hiç farkında olmadığım bir şey ” diyor .

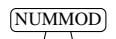

3) *Adverbial Modifiers*: ADV tag marks adverbial modifiers and ADVCL tag marks the clausal adverbial modifiers. Adverbial modifiers modify the predicates or adjectives. As opposed to the Universal Dependencies (UD) v.2 for Turkish, we didn’t use the “advmod:emph” tag for the adverbials that are used for emphasis.

(22)  Marni Rice , hizmetçi çok iyi oynar .

(23) “ Onu şimdi çok daha yakından izliyoruz ” dedi .


4) *DET, NUMMOD*: Determiners and numeral modifiers are the last two types of modifiers. They are marked as DET and NUMMOD respectively. The indefinite articles and demonstratives are marked as DET. NUMMOD is used to mark any type of number-noun relationship. This differentiation and manual tagging made it possible to distinguish the indefinite article “bir” from the number “bir”.

(24)  Bu bir attan düşmek ve tekrar binmek gibi bir şey .

(25)  Sonaçlar bir gün sonra açıklandı .

V. INCONSISTENCIES IN ANNOTATIONS

Just like in any annotation process performed by several annotators, we have also had issues concerning inter-annotator agreement. Even though each annotator follows the same guidelines, certain items can be annotated with different tags.

In some cases, one item is supposed to have different tags. For example, as we have mentioned above, “bir” can function either as an indefinite determiner (DET) or a numeral (NUMMOD). Also, certain items can appear as adjectives (AMOD) when they modify a noun or as adverbs (ADVMOD) when they modify verbs. Items such as “sadece”, “düşük”, “az”, “kötü” are some of the items which were annotated as ADVMOD or AMOD according to their context.

Apart from this kind of inevitable and necessary double taggings, we noted some disagreements on some items like

conjunctions and adjuncts. For instance, whenever the conjunction “ancak” is at the beginning of a sentence, it is tagged with either CC or DISCOURSE. As the previous sentences are not within the scope of our syntactic analysis, “ancak” can’t be shown to link two different utterances. So, both tags can be justified. Lastly, temporal adjuncts like “bugün”, “bu yıl” are sometimes interpreted as adverbials (ADVMOD), or temporal modifiers (NMOD). Such inconsistencies are not eliminated in order to achieve a broader range of learning possibilities for the parser.

VI. CONCLUSION

As the final outcome of this project, we are now in possession of 16.400 sentences that were morphologically, semantically and syntactically annotated by a team of linguists. In addition to being the largest Turkish dependency corpus, it is also the first one which is cross-linguistic. Considering that the sentences of our data were translated from the Penn Treebank corpus, they constitute a treebank which is parallel to its English version, enabling us to observe the structures and rules of the two languages clearly. We hope that this cross-linguistic corpus will be the first of many for Turkish and contribute to the developments in machine translation.

Furthermore, the data compiled within the scope of this project will eventually be used to create an automatic dependency parser for Turkish. In addition, a parallel project which consists of the automatic conversion of English constituency trees into Turkish is currently being carried out. With the help of the phrase structure rules obtained from this project, we are hoping to develop a system that can automatically convert constituency trees into dependency trees and vice versa.

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